

Part-Time RA/PDRA Recruitment

PDRA/RA Position - Resource Allocation in Federated HPC Systems - QMUL

We are recruiting a Postdoctoral Research Associate (PDRA) or Research Assistant (RA) with a PhD/MSc degree in a quantitative field at Queen Mary University of London for the "FAIR-Compute" project. This is a part-time, fixed-term position. Project end date is on 31-Jul-26.

The project, funded by the NFCS NetworkPlus (<https://nfcs-networkplus.ac.uk>), aims to develop a foundational roadmap for the fair and efficient allocation of federated computational resources (e.g., HPC/HTC).

The appointee will lead the technical simulation and modelling work package, responsible for designing and building computational models to evaluate non-monetary resource allocation mechanisms. This is an interdisciplinary role co-supervised by Prof. Kostas Zachariadis (School of Economics and Finance, QMUL) and Dr. Ahmed Sayed (School of Electronic Engineering and Computer Science, QMUL).

Essential Criteria

- Strong demonstrable experience in one or more of: Resource Allocation, Simulation and Modeling, Experimental Design, Market Design (without monetary transfers), or Algorithmic Game Theory.
- Demonstrable programming experience in Python, C++, Java, MATLAB, or an equivalent high-level language.
- Prior experience with simulation-based experiments and Cloud/HPC simulation tools such as SimGrid, CloudSim, CQSim, etc.
- Excellent analytical, computational, communication, research, and problem-solving skills.

Desirable Criteria

- Knowledge of federated computing, cloud computing, or distributed systems.
- Familiarity with the "mechanism design without money" or "fair division" literature.
- Direct experience in building agent-based models or other economic simulations.

For further info please contact Prof. Kostas Zachariadis (k.e.zachariadis@qmul.ac.uk) or Dr. Ahmed Sayed (ahmed.sayed@qmul.ac.uk)